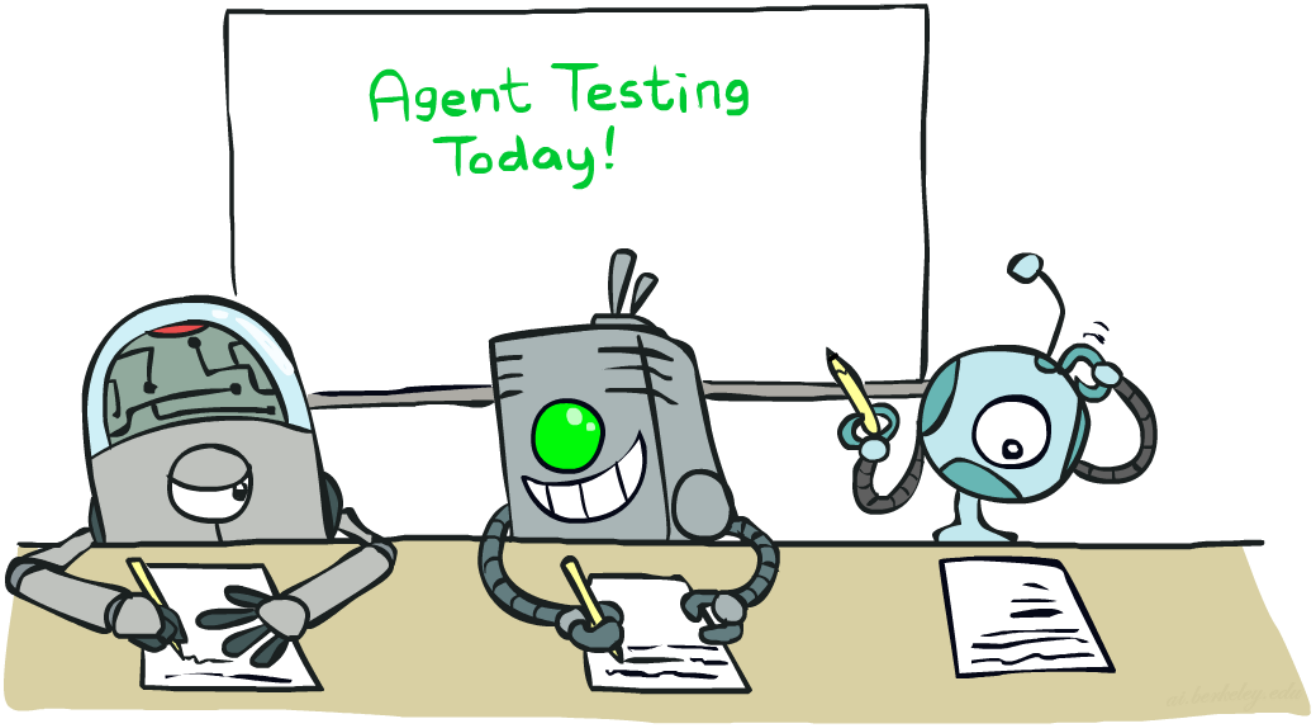
Machine Learning



Machine Learning

- Up until now: how use a model to make optimal decisions

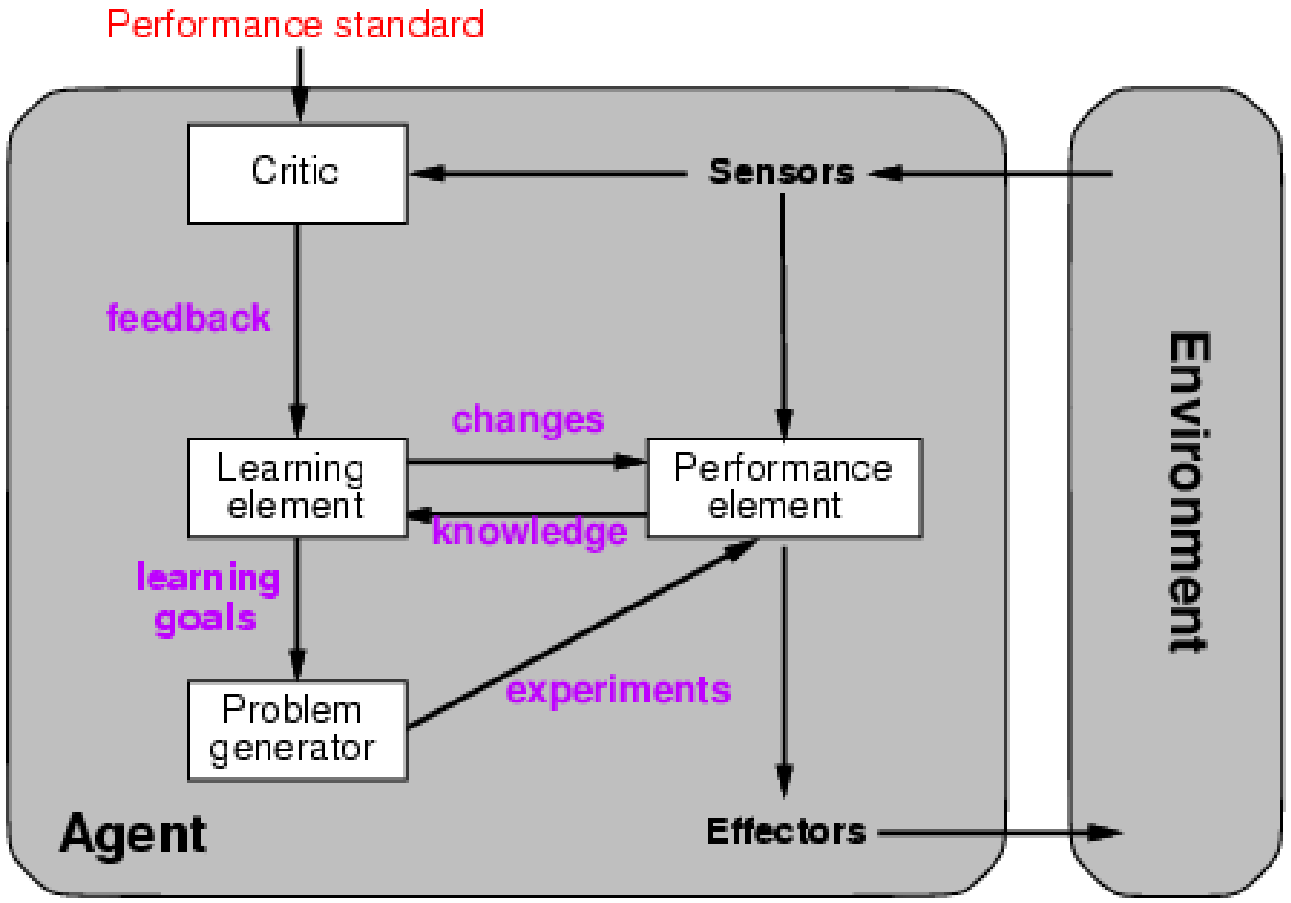
- Machine learning: how to acquire a model from data / experience
  - Learning parameters (e.g. probabilities)
  - Learning structure (e.g. BN graphs)
  - Learning hidden concepts (e.g. clustering, neural nets)

- Today: model-based classification with Naive Bayes

Architecture for Intelligence

- Search?
  - Solve the problem of what to do.
- Learning?
  - Learn what to do.
- Logic and inference?
  - Reason about what to do.
  - Encoded knowledge/”expert” systems?
    - Know what to do.

Learning agents



The Learning Problem

- Learning = Improving with experience at some task
  - Improve over task T
  - With respect to performance measure P
  - Based on experience E
- Example: Learn to play checkers
  - T: Play checkers
  - P: % of games won in world tournament
  - E: opportunity to play against self
- Example: Learn to Diagnose Patients
  - T: Diagnose patients
  - P: Percent of patients correctly diagnosed
  - E: Pre-diagnosed medical histories of patients

Categories of Learning

- Learning by examples / Supervised learning
- Learning by discovery / Unsupervised learning
- Learning by experimentation / Reinforcement learning



Types of learning

- **Supervised learning**: learn mapping from attributes to “target”
  - Classification: target variable is discrete (e.g., spam email)
  - Regression: target variable is real-valued (e.g., stock market)
- **Unsupervised learning**: no target variable; “understand” data structure
  - Clustering: grouping data into K groups
- **Reinforcement learning**: e.g., game-playing agent
  - the agent does not know the exact output for an input, but it receives feedback on the desirability of its behavior
    - the feedback can come from an outside entity, the environment, or the agent itself
    - the feedback may be delayed, and not follow the respective action immediately

Classical Machine Learning

Task Driven

Data Driven

Supervised Learning

(Pre Categorized Data)

Classification

(Divide the socks by Color)

Eg. Identity Fraud Detection

Regression

(Divide the Ties by Length)

Eg. Market Forecasting

Unsupervised Learning

(Unlabelled Data)

Clustering

(Divide by Similarity)

Eg. Targeted Marketing

Association

(Identify Sequences)

Eg. Customer Recommendation

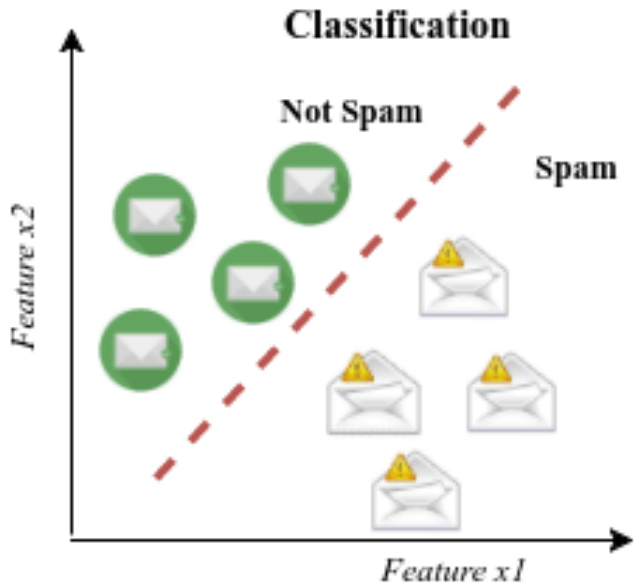
Dimensionality Reduction

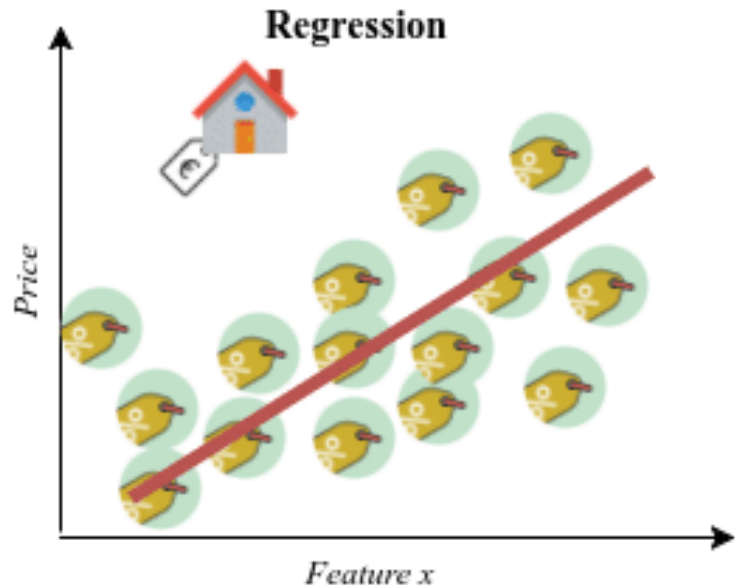
(Wider Dependencies)

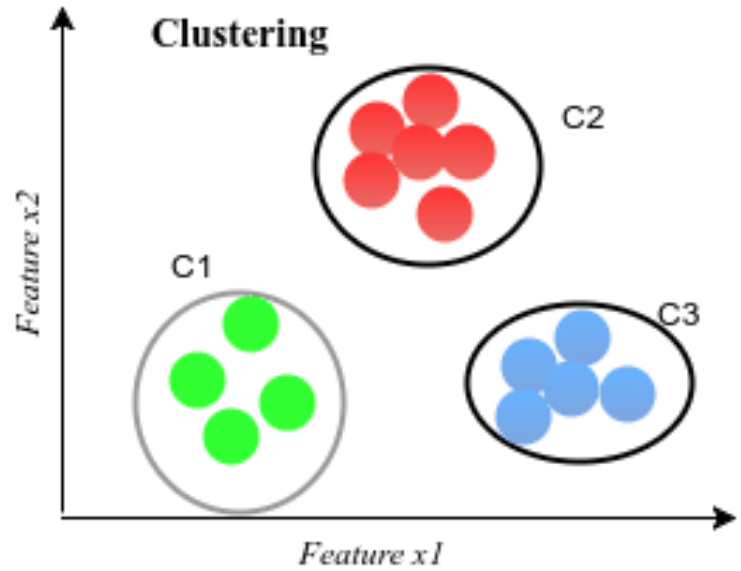
Eg. Big Data Visualization

Obj: Predications & Predictive Models

Pattern/ Structure Recognition


Classification
Not Spam
Spam
Feature x2
Feature x1

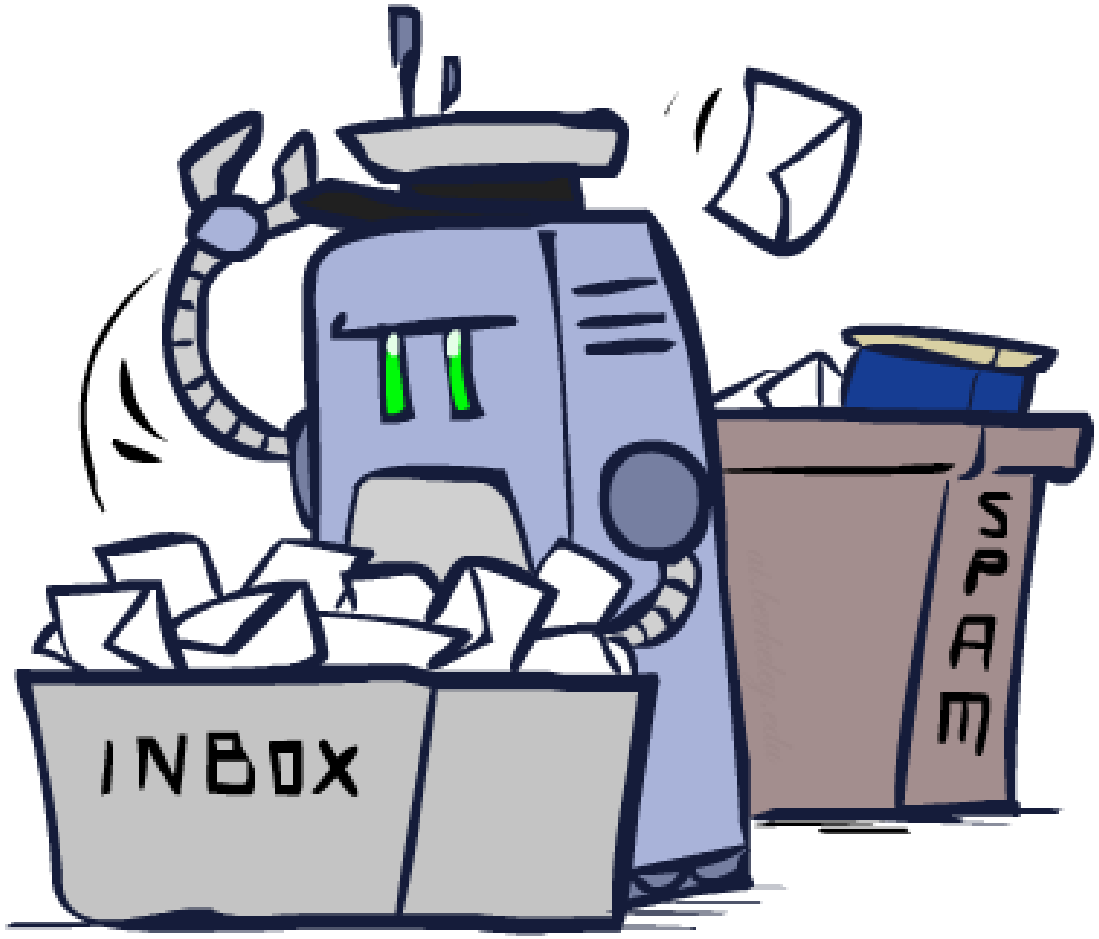

Regression
Price
Feature x


Clustering
C1
C2
C3
Feature x2
Feature x1

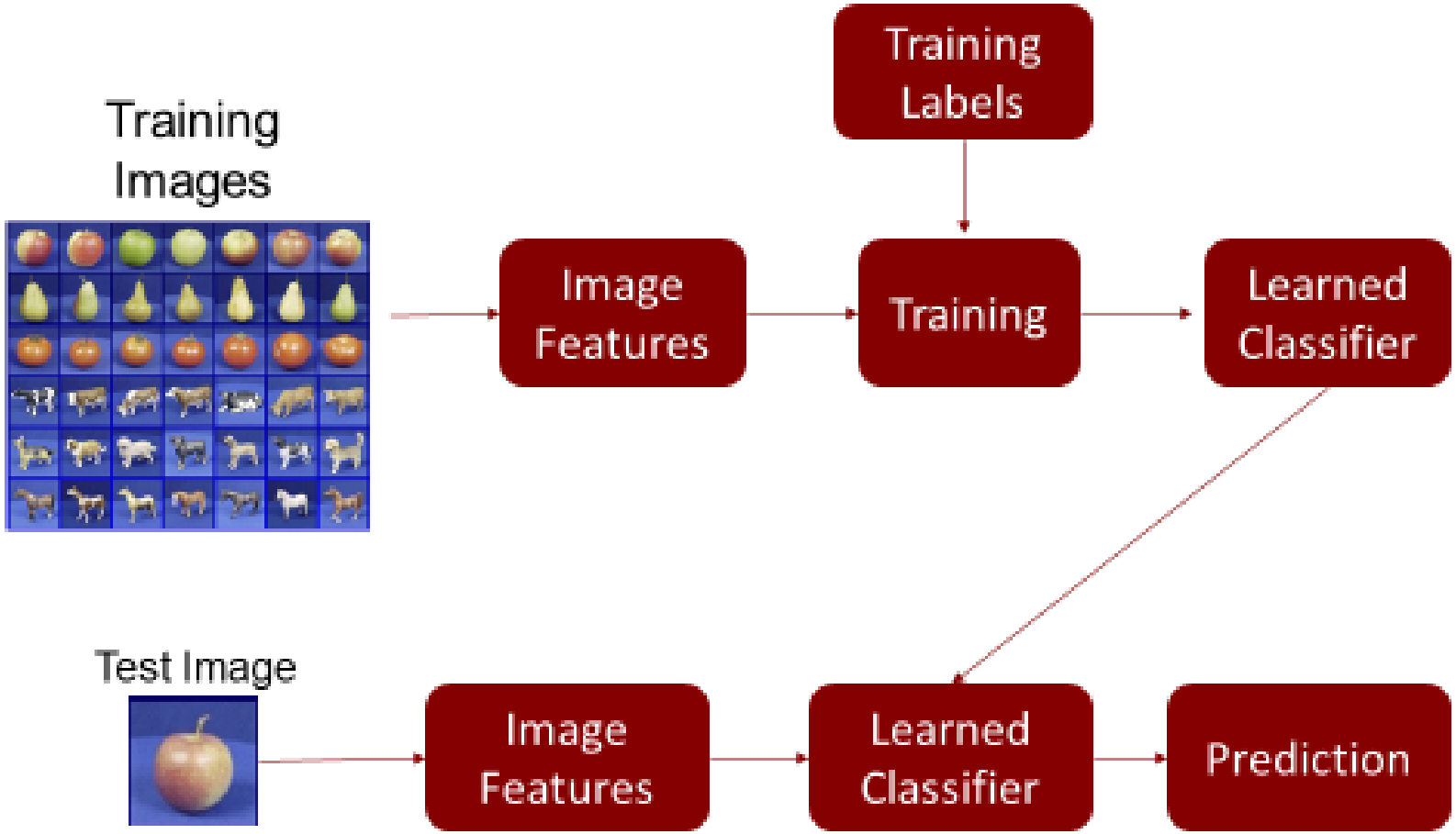
Learning From Examples

- Learn general concepts or categories from examples
- Learn a task (drive a vehicle, win a game of backgammon)
- Examples of objects or tasks are gathered and stored in a database
- Each example is described by a set of attributes or features
- Each example used for training is classified with its correct label (chair, not chair, horse, not horse, 1 vs. 2 vs. 3, good move, bad move, etc.)
- The machine learning program learns general concept description from these specific examples
- The ML program should be applied to classify or perform tasks *never before seen* from learned concept

Classification



A simple pipeline - Training



Example: Spam Filter

- Input: an email
- Output: spam/ham

- Setup:
  - Get a large collection of example emails, each labeled “spam” or “ham”
  - Note: someone has to hand label all this data!
  - Want to learn to predict labels of new, future emails

- Features: The attributes used to make the ham / spam decision
  - Words: FREE!
  - Text Patterns: $dd, CAPS
  - Non-text: SenderInContacts, WidelyBroadcast
  - …



Dear Sir.

First, I must solicit your confidence in this transaction, this is by virture of its nature as being utterly confidencial and top secret. …



TO BE REMOVED FROM FUTURE MAILINGS, SIMPLY REPLY TO THIS MESSAGE AND PUT "REMOVE" IN THE SUBJECT.

99 MILLION EMAIL ADDRESSES FOR ONLY $99



Ok, Iknow this is blatantly OT but I'm beginning to go insane. Had an old Dell Dimension XPS sitting in the corner and decided to put it to use, I know it was working pre being stuck in the corner, but when I plugged it in, hit the power nothing happened.

Example: Digit Recognition

- Input: images / pixel grids
- Output: a digit 0-9

- Setup:
  - Get a large collection of example images, each labeled with a digit
  - Note: someone has to hand label all this data!
  - Want to learn to predict labels of new, future digit images

- Features: The attributes used to make the digit decision
  - Pixels: (6,8)=ON
  - Shape Patterns: NumComponents, AspectRatio, NumLoops
  - …
  - Features are increasingly induced rather than crafted


0


1


2

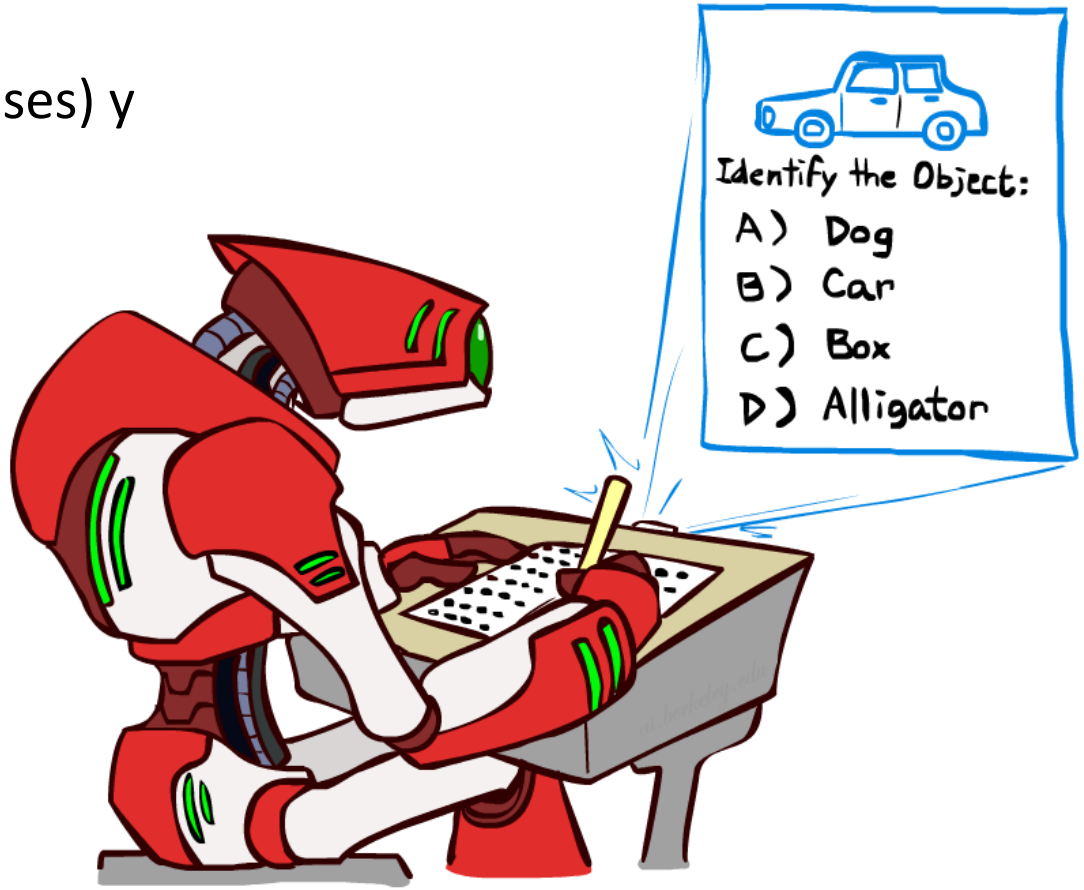

1


??

Other Classification Tasks

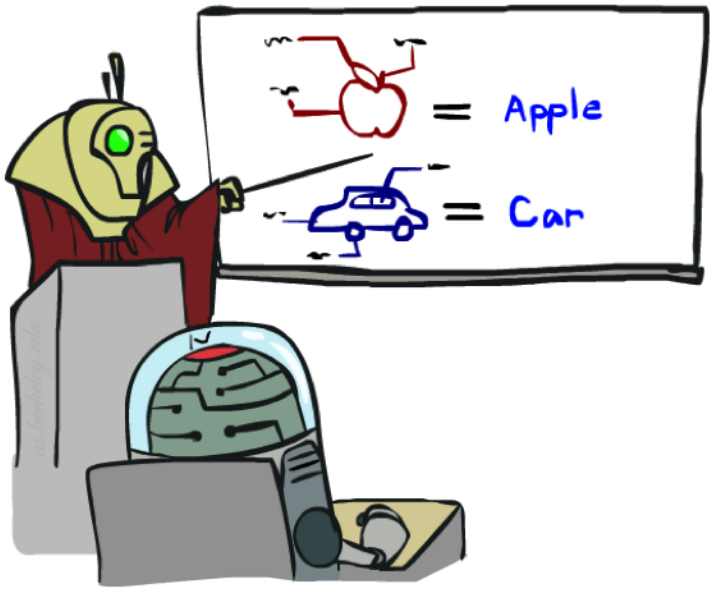
- Classification: given inputs x, predict labels (classes) y

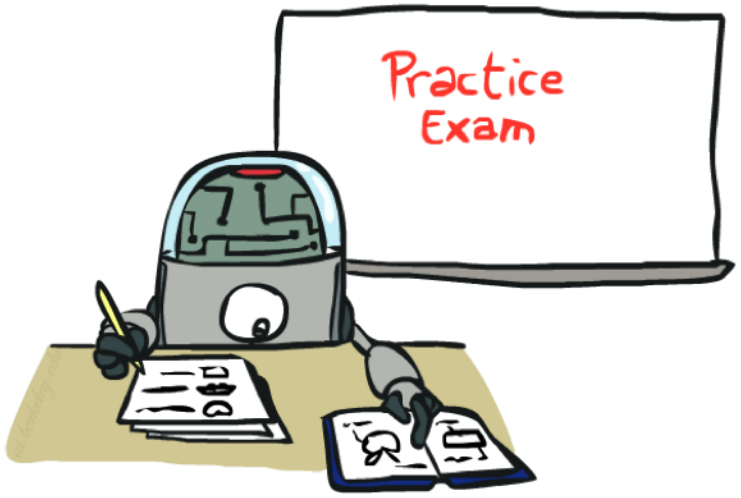
- Examples:
  - Medical diagnosis (input: symptoms, classes: diseases)
  - Fraud detection (input: account activity, classes: fraud / no fraud)
  - Automatic essay grading (input: document, classes: grades)
  - Customer service email routing
  - Review sentiment
  - Language ID
  - … many more

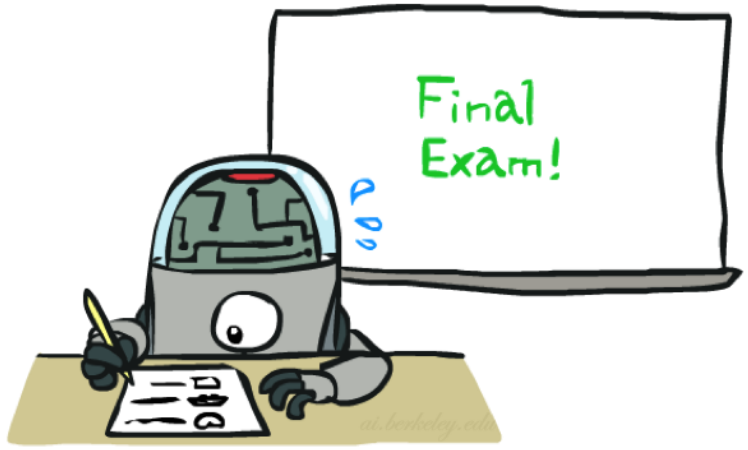


- Classification is an important commercial technology!

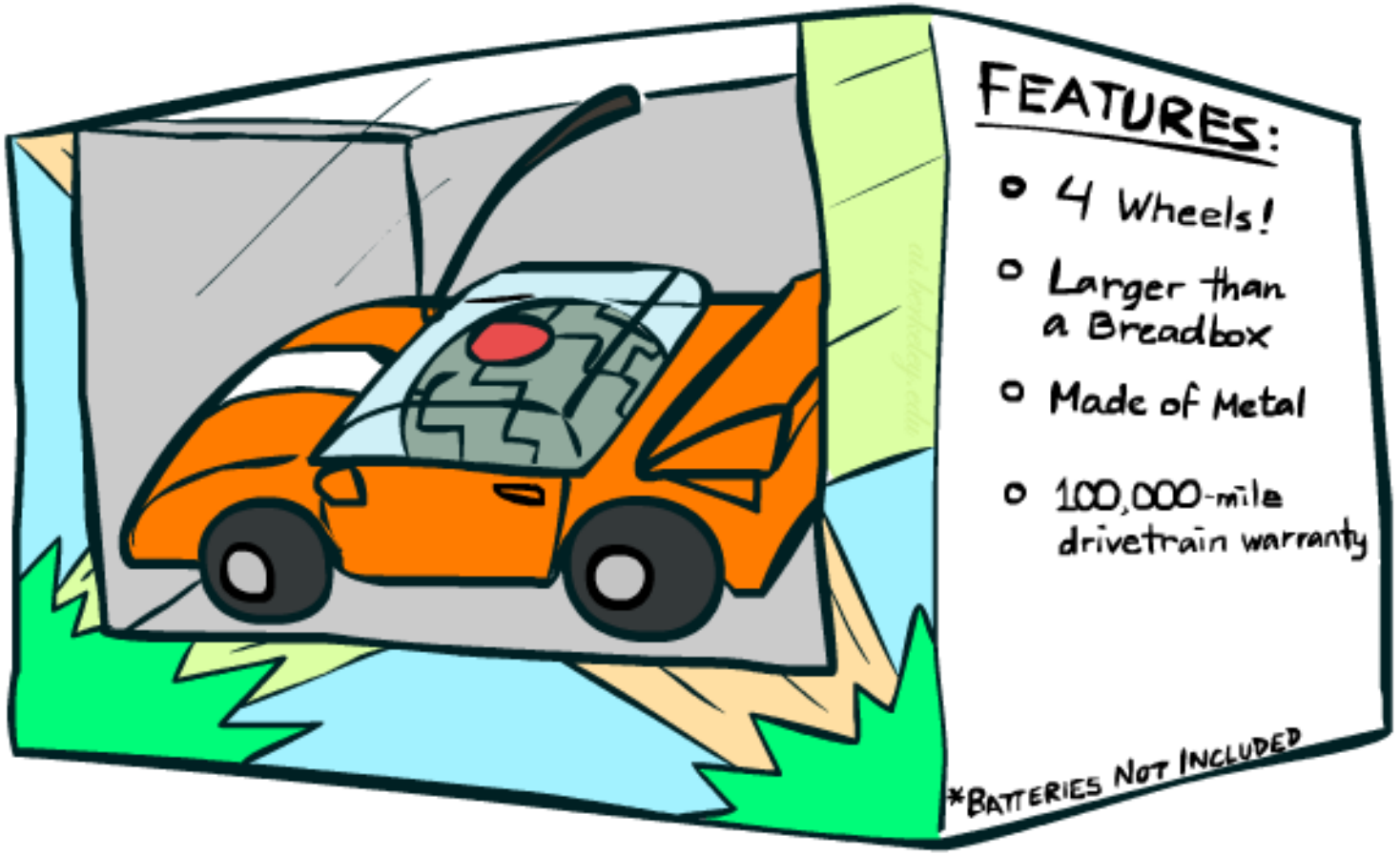
Training and Testing







Features


FEATURES:
4 Wheels!
Larger than a Breadbox
Made of Metal
100,000-mile drivetrain warranty
*Batteries Not Included

Illustrative Example

Target Label Y

Feature vector

Day	*Sunny*	*Temp.*	*Humidity*	*Wind*	*PlayTennis*
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Another example

Features

Target Label Y

Size (feet^2) x1	Number of bedrooms x2	Number of floors x3	Age of home (years) x4	Price ($1000)
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
…	…	…	…	…

Movie review data example

From text to vectors

$$V = [v_1, v_2, v_3, \cdots, v_n, class]$$

review$_1$="great movie"

review$_2$="excellent film"

review$_3$="worst film ever"

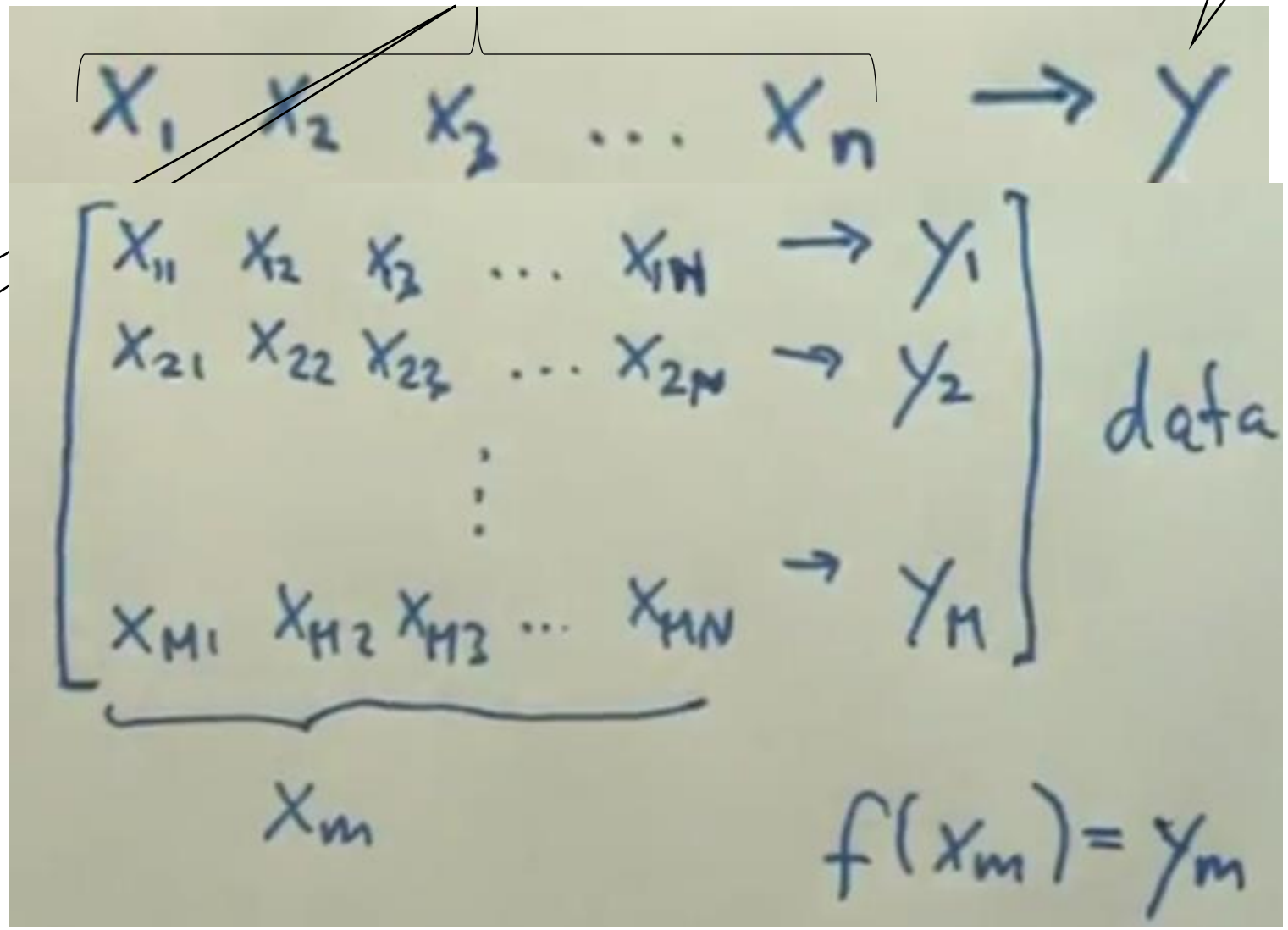
review$_4$="sucks"

Features

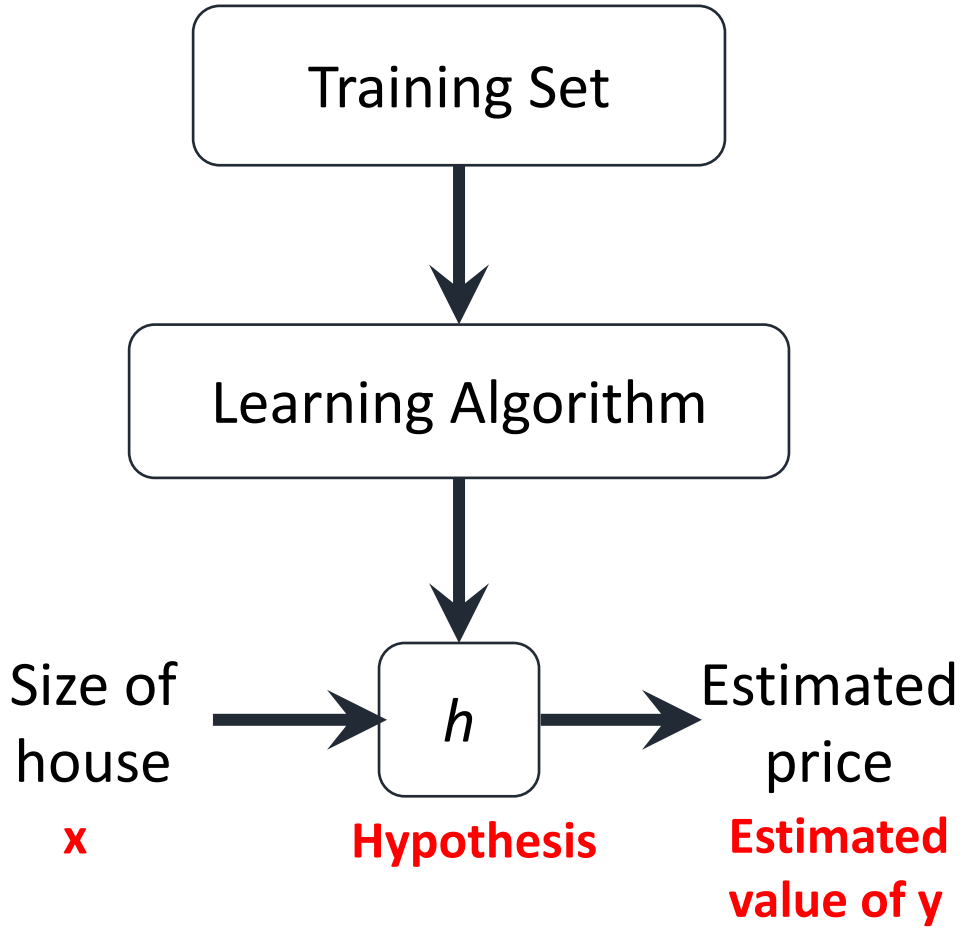
ever	excellent	film	great	movie	sucks	worst	
0	0	0	1	1	0	0	+
0	1	1	0	0	0	0	+
1	0	1	0	0	0	1	−
0	0	0	0	0	1	0	−

Target Label Y

Supervised Learning



Supervised Learning



h is a function

h maps from x's to y's